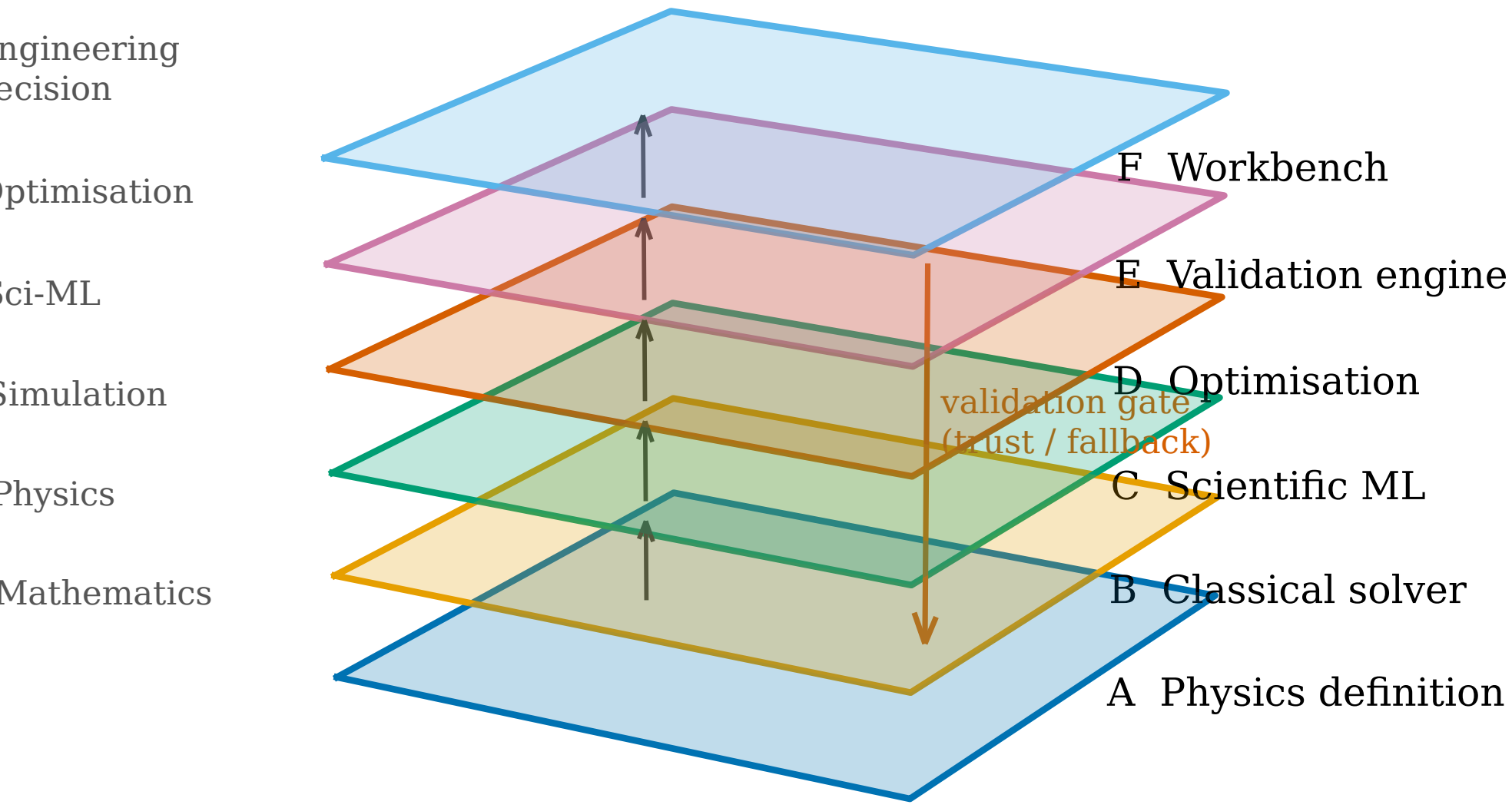Schematic (not data): layered architecture and research pipeline